

GroupTrack V2.4 — Release Notes & User Guide

Release date: May 2026 **Audience:** GroupTrack testers and field users **Distribution:** Play Store internal testing track + sideload APK

TL;DR — What's New

GroupTrack V2.4 is the first Play Store-ready release. Major changes:

1. **Work With Tracks** — full in-app track management (rename, delete, share, move, fix dates)
 2. **Phone GPS fallback** — works without a radio in phone-only mode
 3. **Trailer cart substitution** — silent carts shown at trailing cart position so they're never "lost on map"
 4. **Reliable autopause** — battery savings during stops, automatic wake on movement
 5. **Stable track recording** — RECORD button responds instantly, no flickering, no crashes on track load
 6. **Date-aware track library** — sort by actual recording date, not import date
 7. **Map Viewer overhaul** — manage tracks directly from the map without leaving viewer
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New Features — How to Use

1. Work With Tracks (Main Menu → Export Tracks to Downloads)

Replaces the old file-manager launcher with a full in-app track manager.

What you can do: - **Browse all tracks** in your `my_tracks/` folder - **Search** by typing — list filters live as you type - **Sort** by tapping the date/name button: cycles Date↓ → Date↑ → Name↑ → Name↓ - **Filter tabs:** - *All* — everything - *Saved* — completed tracks (default usable view) - *In-Progress* — orphaned `convoy_track_temp_*` files from interrupted recordings (review and delete manually) - **Per-track actions** (tap any row): Rename, Delete, Share (email/drive), Move to Downloads, Fix Creation Date - **Bulk Fix Dates** — button next to track count applies date-fix to all visible tracks at once

No auto-delete: GroupTrack never deletes your tracks automatically. Cleanup of orphan in-progress files is a deliberate user action.

2. Track Date Preservation

Your tracks now show their **actual recording date** instead of import date.

GroupTrack reads the `<time>` elements inside each GPX/KML file and uses the earliest one as the file's date. Sort by Date and you'll see tracks in true chronological order — even tracks imported from Gaia, onX, or other tools years after they were recorded.

For tracks imported before V2.4: their dates currently show the import date. To fix: -

Single track → tap the track → action menu → **Fix Creation Date - All visible tracks** → tap the **Fix Dates** button at the top of Work With Tracks

This rewrites only the file's metadata timestamp — the actual GPS data inside the file is untouched.

3. Map Viewer Track Actions

Same management actions are now available directly in Map Viewer.

- Open Map Viewer
- Tap a track in the list to load/unload it on the map (existing behavior — unchanged)
- **Tap the ! icon** on any track row to open the action menu: Rename, Delete, Share, Move to Downloads, Fix Creation Date

Smart handling: if you rename or delete a track that's currently displayed on the map, GroupTrack automatically removes it from the map first.

4. Phone GPS Fallback (No Radio? Still Works.)

If GroupTrack can't reach a radio (no Bluetooth, radio off, or you're using phone-only mode), the app now falls back to the phone's own GPS for your cart position.

Order of priority: 1. Radio GPS (preferred, accurate, no battery hit) 2. Last known radio position (if radio went silent recently) 3. Phone GPS (final fallback)

You'll always see your cart on the map, regardless of radio state.

5. Trailer Cart Substitution

When a cart in the middle of your convoy stops communicating (battery, BLE drop, weak mesh signal), GroupTrack now shows that silent cart at **at least the position of the trailing cart** that's still reporting.

Why: the silent cart is somewhere between you and the trailing cart. Showing it at the trailing position keeps it visible on the map and prevents it from "disappearing back to its last known position 5 miles ago."

Off-track carts are exempt — if a cart deviated from the convoy track on purpose, its own last-known position is preserved so the operator can find it.

6. Autopause + Auto-Wake (Restored & Improved)

When the convoy stops moving for ~10 minutes within ~50 feet of one location, GroupTrack: - Pauses GPS updates - Disconnects from radio (saves battery) - Shows a flashing "ZZZ" indicator - Wakes automatically when the operator taps the wake icon (reconnects radio, resumes GPS)

Use case: Long lunch breaks, fuel stops, overnight camp. No battery drain, instant resume.

7. RECORD Button Bounce Fix

Tapping RECORD now **immediately centers the map on your cart**. Previously there was a 1-3 second gap before the next position update painted your cart on the map. No more visual lag.

8. Track Selection No Longer Crashes

Loading a large GPX track on either the Convoy screen or Map Viewer used to occasionally cause an Application Not Responding (ANR) crash. This is now fixed — file reading and parsing happen on a background thread.

Stability & Performance Fixes (May 2-7, 2026)

These are the under-the-hood fixes that make V2.4 ship-ready:

Fix	Why it matters
Removed blocking permission request from app start	Fresh installs no longer ANR on first launch
All-files storage permission dialog moved into Convoy screen	Permission asked at the right time, not before app is usable
Permission re-prompt on app resume	If you revoke permissions in Settings, the app prompts again on next return
Identity locked at RECORD time	Your cart ID stops flickering during convoy joins
AutoPan default off	Map no longer "snaps back" when you're not actively recording
WebView cleanup on Map Viewer exit	No memory leak from repeated map opens
On-demand trail loading	Trails only load when you turn them on, faster startup
z18 bypass	Map zoom feels smoother
SHOW DOWNLOADED progress feedback	You see download status while caching offline tiles
SHA-256 fingerprints registered in Firebase	Production package <code>com.grouptrack.android</code> works with Google services

Known Limitations (Acknowledged for V2.5)

- **Import flash:** when you tap a KML/GPX in Android's file picker (e.g., from Gmail or Drive), there's a brief grey/black screen flash before returning to GroupTrack. The import works correctly — flash is cosmetic. V2.5 will add an in-app Downloads browser to bypass this.
- **Imported track names with timestamps:** GroupTrack-recorded tracks have `_YYYYMMDD_HHMMSS` suffix. V2.5 will use location-based naming.
- **Esri online maps:** sparse labels online vs dense offline. This is Esri's deprecation of legacy services, not a GroupTrack bug. Your offline maps remain rich. V2.5 will plan migration.
- **Tab8NEU and budget Android devices** (Android 10/14/16): cannot maintain BLE connection to T1000-E radios — supervision timeout after 8-10s. Use stable devices (Android 1, Android 2 verified working).

User Manual — Sections to Update or Add

Section: "Managing Your Tracks" (NEW SECTION)

Replace any existing "Export Tracks" or "Track Library" section with the following:

Managing Your Tracks

GroupTrack keeps all your imported and recorded tracks in `Documents/my_tracks/` on your device. To manage them, open the convoy menu and tap **Export Tracks to Downloads** (this opens the new Work With Tracks screen).

Searching: Type into the search box at the top. The list filters live as you type.

Sorting: Tap the date/name button to cycle through four sort modes — Date Newest, Date Oldest, Name A-Z, Name Z-A. Default is Date Newest.

Filtering: Three tabs: - *All* — every file in your tracks folder - *Saved* — your completed tracks (the normal view) - *In-Progress* — abandoned recording files from interrupted sessions; review and delete manually

Track actions: Tap any track to open its action menu: - **Rename** — change the file name; the .gpx or .kml extension is preserved automatically - **Delete** — permanent removal; confirms before acting - **Share** — opens Android's share sheet, attach to email, send to Drive, AirDrop equivalents, etc. - **Move to Downloads** — copies the file to your Downloads folder for backup or transfer; the original stays in my_tracks - **Fix Creation Date** — updates the file's date to match the earliest GPS timestamp inside the file

Bulk fix dates: The **Fix Dates** button next to the track count rewrites date stamps for every visible track at once. Use this once to clean up tracks imported before V2.4 — they'll start showing their real recording dates.

Section: "Map Viewer" — ADD TO EXISTING

Add the following at the end of your Map Viewer documentation:

Track Actions in Map Viewer

Each track in the Map Viewer track list now has an overflow icon () on the right side. Tapping the icon opens the same action menu available in Work With Tracks: Rename, Delete, Share, Move to Downloads, Fix Creation Date.

Tapping the track name itself still loads (or unloads) the track on the map — that behavior is unchanged.

If you rename or delete a track that's currently displayed on the map, GroupTrack automatically removes it from the map first to prevent stale references.

Section: "Cart Position & Tracking" (UPDATE)

Add or replace your current text on cart positioning with this:

Cart Position & Tracking

GroupTrack determines each cart's position with a four-tier fallback chain:

1. **Live radio GPS** — preferred. The cart's radio reports its current GPS fix.
2. **Last known radio position** — if the radio is currently silent but reported a position recently, that position is preserved.
3. **Phone GPS** — for the user's own cart, if the radio cannot be reached, the phone's GPS is used.
4. **Trailer cart substitution** — for on-track carts that have gone silent, the cart is shown at the position of the furthest trailing cart that is still reporting. This keeps silent carts visible on the map and prevents them from appearing to "fall behind" indefinitely.

Carts that have deviated off the planned track (off-track status) keep their own last known position so operators can locate them.

Section: "AutoPause & Auto-Wake" (NEW SECTION)

Add this section if not already present:

AutoPause & Auto-Wake

When your convoy is stationary for approximately 10 minutes within roughly 50 feet of one spot, GroupTrack automatically pauses to save battery:

- GPS updates pause
- Radio Bluetooth disconnects
- The screen displays a flashing "ZZZ" indicator

This is ideal for fuel stops, lunches, and overnight camps.

To wake: tap the ZZZ indicator. GroupTrack reconnects to your radio, resumes GPS, and rejoins the convoy.

Section: "Recording Tracks" (UPDATE)

Add the following note:

When you tap **RECORD**, the map immediately centers on your cart's position. This is a change from earlier versions where there was a brief delay before the cart appeared centered.

While recording, the AutoPan feature stays on — the map follows your cart. When you stop recording, AutoPan turns off so you can pan and zoom freely.

For Testers — How to Validate V2.4

Track manager: - [] Open convoy menu → Export Tracks to Downloads → confirm Work With Tracks opens - [] Type a search term → list filters - [] Tap sort button several times → cycles through 4 modes - [] Tap each filter tab → confirm correct subset shows - [] Tap a track → action sheet appears - [] Rename a test track → confirm change persists - [] Delete a test track → confirm gone after refresh - [] Share a track via email → file attached - [] Move a track to Downloads → file appears in Downloads - [] Fix date on a 2026-05-01 13:51 track → date updates to actual recording date

Map Viewer: - [] Open Map Viewer → tap a track → loads on map (existing behavior) - [] Tap ⋮ icon on a loaded track → action menu appears - [] Rename loaded track → automatically unloads from map first - [] Delete loaded track → unloads then deletes

Recording: - [] Tap RECORD → map immediately centers on your cart (no 1-3s gap) - [] Stand still for 10 minutes → ZZZ flashing indicator appears - [] Tap ZZZ → reconnects, resumes recording

Permissions: - [] Fresh install → no ANR on first launch - [] Revoke storage permission in Settings → app prompts on next resume - [] Grant all-files access through dialog → tracks load

Document version: V2.4 ship release, May 7 2026

What's Next — V2.5 Coming Soon

This is the last release for approximately 4 weeks. The next dev cycle adds the following V2.5 functionality:

In-app Downloads import

A new "Import from Downloads" screen mirroring Work With Tracks, pointed at your Downloads folder. Browse and import GPX/KML files without leaving GroupTrack — eliminates the screen flash that happens when opening a file from Android's file picker.

Multi-select with bulk actions

Long-press any track in Work With Tracks or Map Viewer to enter multi-select mode. Select multiple tracks with checkboxes, then apply bulk operations: - Delete multiple tracks at once - Share multiple tracks (multi-attach email) - Move multiple tracks to Downloads - Fix Creation Date on selected tracks - Select All / Deselect All - Rename excluded (each rename needs unique input)

Smarter track naming

GroupTrack-recorded tracks will get meaningful names based on starting location instead of timestamp suffixes. Imagine `Coral_Pink_Sand_Dunes.gpx` instead of `coral_pink_sand_day_2_20260505_155359.gpx`.

Esri map service migration planning

Esri's legacy ArcGIS Online services (the ones GroupTrack uses for satellite, topo, and labels) are being sunsetted. V2.5 will plan the migration to Esri's new authenticated basemap service or evaluate alternatives (Google Maps SDK with native MapView for unlimited free tiles).

Cache versioning + "Refresh Tiles" option

Track which URL/version your map tiles came from. When map sources change, refresh affected regions instead of reloading every region.

Tile interceptor logging

Improved debug logging when investigating map issues — currently only logs `convoy://` URLs, will add label tile fetches.

Cleanup automation for orphan tracks

The `In-Progress` filter in Work With Tracks already isolates orphaned `convoy_track_temp_*` files. V2.5 will add a "Delete All In-Progress" button (with confirm + size totals) — still user-initiated, never automatic.

Auto-clear launcher cache on install

Capture the launcher icon duplicate fix into the install procedure so testers don't have to manually clear the launcher cache.

Native Android GPS speed

Drop the 60-second windowed speed calculation in favor of Android's built-in `Location.getSpeed()` — faster speed updates, less latency.

Fix import flash (manifest intent flow)

The grey/black screen flash when opening KML/GPX from another app is on the V2.5 list — investigated but not fixed in V2.4.

Why the 4-week pause?

The V2.4 build represents a stable baseline. Pushing changes during the upcoming travel period would risk introducing regressions without test coverage. The pause lets V2.4 settle in tester hands, gather field feedback, and start V2.5 with a clean slate.

Tester feedback during the pause is invaluable. Use the V2.4 features extensively. Note anything that surprises you, frustrates you, or is missing. That feedback drives V2.5 priorities when development resumes.

GroupTrack thanks all V2.4 testers — your field reports during the April-May ride season made this release possible.